

Code Explanation

Code Explanation: Package Initialization File

Overview

This is a Python package initialization file (`\_\_init\_\_.py`) for an AWS Glue PySpark ETL job that processes customer and order data.---

Step-by-Step Breakdown

1. **Module Docstring (Lines 1-9)**

```
"""AWS Glue PySpark ETL Job - Customer Order Analytics with SCD Type 2..."""
```

- Describes the package purpose: ETL job for customer/order analytics- Key technology: **SCD Type 2** (Slowly Changing Dimension Type 2) for tracking historical changes- Storage format: **Apache Hudi** (enables incremental data processing and time-travel queries)- Lists contained modules: ``job`` module contains main ETL logic

2. **Package Metadata (Lines 11-13)**

```
__version__ = "1.0.0"__author__ = "Data Engineering Team"__description__ =  
"Customer Order Analytics ETL with SCD Type 2..."
```

- Defines version, authorship, and description- Standard Python convention for package identification

3. **Public API Declaration (Lines 15-18)**

```
__all__ = ["job"]
```

- Controls what gets exported when someone does ``from package import *``- Only the ``job`` module is exposed as the public interface- Prevents accidental import of internal/helper modules---

Purpose

This initialization file:- Documents the package's role in the ETL pipeline- Establishes version control metadata- Defines the public API surface (only ``job`` module)- Serves as entry point documentation for developers---

Key Concepts Referenced

- **SCD Type 2**: Maintains full history by creating new records for changes (vs. updating in place)- **Apache Hudi**: Data lake storage format supporting ACID transactions and incremental processing- **AWS Glue**: Managed ETL service running PySpark jobs